

MLOPS IN PRODUCTION · ISSUE 04 OF 08

Monitoring signals and retraining triggers.

Automate retraining. Gate promotion.

04

AR

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Swipe →

Signals to track

01

Service: latency, errors

02

**Data: schema, nulls,
freshness**

03

**Model: prediction
distribution**

04

Model: confidence

05

**Outcome: labeled
accuracy**

06

**Business: conversion,
cost**

07

Pipeline: job success, lag

08

Features: train/serve skew

Retraining triggers

TRIGGER	FITS	RISK
Schedule	Stable domains	Wasted compute
Drift threshold	Fast-moving inputs	Noisy if untuned
Performance drop	When labels arrive	Delayed
New data volume	Growing datasets	Arbitrary cut-off

Automated retraining, gated

- 1 Trigger
- 2 Validate new data
- 3 Train
- 4 Evaluate vs champion
- 5 Approve
- 6 Canary

Validate the data before training. Retraining on corrupted data automates the outage.

Prediction shift alert

```
YAML
- alert: ModelPositiveRateShift
  expr: |
    abs(
      avg_over_time(model_positive_rate[1h])
      - avg_over_time(model_positive_rate[7d])
    ) > 0.10
  for: 30m
  labels: { severity: ticket }
```

A ticket, not a page. Drift needs a human look, not a 3am wake-up.

Retraining rules

DO

- ✓ Auto-retrain on a trigger
- ✓ Compare against the champion
- ✓ Keep every candidate in the registry

AVOID

- ✗ Auto-promote without a gate
- ✗ Train on unvalidated data
- ✗ Page humans for drift

KEY TAKEAWAY

Monitor the model like a service and the data like a dependency.

- Add data, model and outcome signals
- Pick triggers to match the domain
- Never auto-promote

YOUR TURN

Which retraining trigger does your team actually use?

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Next in the series: #05 Reproducible pipelines with MLflow and Kubeflow

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